OWNERSHIPGRAPH AI

Evidence-led investigation report

Investigation #1 | Generated 2026-11-07T16:02:05.038837+00:00

SYNTHETIC DEMONSTRATION RECORDS

1. Investigation overview

Aster Maritime · ownership review | Status: In Progress
Aster Maritime Group: 2 upstream persons discovered; observed-indicator score 31.5/100 (Medium).

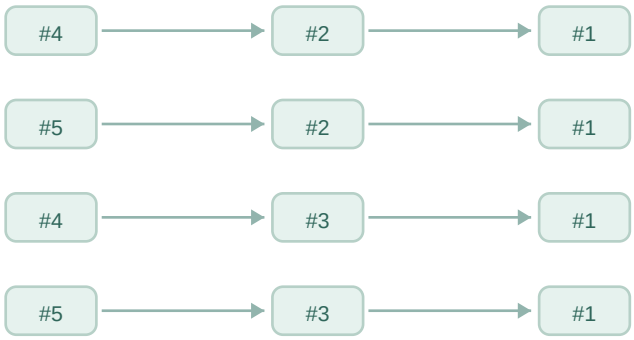
2. Target entity

Aster Maritime Group | COMPANY | United Kingdom | SYN-10001 | SRC-1

3. Ownership structure

4 upstream person paths. Aggregation withheld: True. Threshold: 25% (illustrative).

Ownership path overview (entity IDs; evidence listed below)



4. Potential beneficial owners

Marcus Chen: combined withheld; simple-path estimate 48.0%; candidate: False
Elena Vale: combined withheld; simple-path estimate 42.0%; candidate: False

5. Direct ownership paths

Entity #2 > #1: 60.0% | REL-1 / SRC-1
Entity #3 > #1: 30.0% | REL-2 / SRC-1
Entity #6 > #1: unknown percentage | REL-3 / SRC-1

6. Indirect ownership paths

Entity IDs 4 > 2 > 1: 36.0% | REL-1 / SRC-1, REL-4 / SRC-1
Entity IDs 5 > 2 > 1: 24.0% | REL-1 / SRC-1, REL-5 / SRC-1
Entity IDs 4 > 3 > 1: 6.0% | REL-2 / SRC-1, REL-6 / SRC-1
Entity IDs 5 > 3 > 1: 24.0% | REL-2 / SRC-1, REL-7 / SRC-1

7. Graph findings

Circular ownership: A directed ownership cycle is present. Cross-holdings need reconciliation. [REL-9, REL-10]
Shared director: Shared link across 3 records; this may have a legitimate explanation. [REL-11 / SRC-1, REL-12 / SRC-1, REL-13 / SRC-1]
Common address: Shared link across 2 records; this may have a legitimate explanation. [REL-14 / SRC-1, REL-15 / SRC-1]
Ownership Gap: 10.0% of the share register is unaccounted for or unspecified [REL-1, REL-2, REL-3]
Missing Percentage: Economic ownership percentage is unknown [REL-3]
Unresolved Terminal: No upstream natural-person ownership record is available [ENTITY-6 / SRC-1]

8. Risk assessment

Observed score 31.5/100 | Medium | 60.0% indicator coverage. Partial observed-indicator score. Unavailable indicators contribute zero without weight renormalization; a low score does not establish low risk.

Indicator	Score	Weight	Points
opacity	75.0	20.0%	15.0
complexity	20.0	15.0%	3.0
jurisdiction	Unavailable	15.0%	0.0
screening	Unavailable	20.0%	0.0
network	90.0	15.0%	13.5
inconsistency	0.0	10.0%	0.0
changes	Unavailable	5.0%	0.0

9. Experimental ML output

Status: experimental | Model version: synthetic-isolation-1.0
Novelty score: 0.5791313689481191
Isolation Forest novelty score, not a probability of criminal activity. Feature values are descriptive inputs, not causal attributions.
Inputs: depth=2, shareholders=2, cycles=1, shared_connections=2, concentration=0.45, missing_records=3, cross_jurisdiction_links=9
Training data: synthetic

10. Evidence references

REL-1 / SRC-1
REL-2 / SRC-1
REL-3 / SRC-1
REL-4 / SRC-1
REL-5 / SRC-1
REL-6 / SRC-1
REL-7 / SRC-1
REL-11 / SRC-1

11. Data limitations

No maintained dated jurisdiction reference connected.
Sanctions, PEP and adverse-media sources are not connected.
No validated historical comparison baseline available.
10.0% of the share register is unaccounted for or unspecified
Economic ownership percentage is unknown
No upstream natural-person ownership record is available

12. Recommendations

Request current shareholder registers and corroborating source documents.
Reconcile unknown percentages and upstream legal-entity owners.
Obtain analyst review before any adverse or external decision.

13. Analyst judgments

No analyst notes recorded.

14. Reviewer decision

Reviewer ID: 3 | Decision: Pending / not reviewed | Reviewed at: Not reviewed

15. Responsible use

Indicators support investigation; they are not evidence of criminal activity or a probability of wrongdoing. Synthetic demo data must not support real-world decisions. Human review is required. No sanctions, PEP or adverse-media screening is connected.